

V6355 LCDC DATA SHEET

1. Name: V6355
2. Function: LCD/CRT Display Controller
3. Purpose: Controls screen display
4. Process: YAMAHA Si-Gate CMOS process
5. External
Appearance: 100-pin plastic QFP
6. Electrical Characteristics

(A) Absolute Maximum Ratings

Parameter	Symbol	Minimum	Maximum	Unit
DC Supply Voltage	VDD	-0.5	+7.0	V
DC Input Voltage	VI	-0.5	VDD+0.5	V
DC Output Voltage	VO	-0.5	VDD+0.5	V
Operating Temperature	TOP	0	+70	°C
Storage Temperature	TSTG	-50	+125	°C

(VSS and AVSS are based on 0.0 V)

(B) Recommended Operating Conditions

Parameter	Symbol	Minimum	Typical	Maximum	Unit
DC Supply Voltage	VDD	4.75	5.0	5.25	V
Operating Temperature	TOP	0	25	70	°C

(VDD and AVDD are the same voltage, and
VSS and AVSS are used as 0.0 V)

(C) D.C. Characteristics

Parameter	Symbol	Condition	Min.	Max.	Unit
High-Level Output Voltage (TTL)	V _{OH}	I _{OH} = -0.4mA	2.7		V
Low-Level Output Voltage (TTL)	V _{OL}	I _{OL} = 0.8mA		0.4	V
High-Level Output Voltage (CMOS)	V _{OH}	I _{OH} < 1μA	V _{DD} -0.4		V
Low-Level Output Voltage (CMOS)	V _{OL}	I _{OL} < 1μA		0.4	V
High-Level Input Voltage	V _{IH}		2.2		V
Low-Level Input Voltage	V _{IL}			0.8	V
High-Level Output Current	I _{OH}	V _{OH} = 2.7 V	-0.4		mA
Low-Level Output Current	I _{OL}	V _{OL} = 0.4 V	0.8		mA
Input Leak Current	I _L		-10	10	μA
Output Leak Current (Tristate)	I _{LZ}		-10	10	μA
Power Supply Current (Operating)	I _{DD}	RL = 5.6Kohm		70	mA
Power Supply Current (Quiescent)	I _{DD}	RL = 5.6Kohm		50	mA
High-Level Clock Input Voltage	V _{CH}		3.6		V
Low-Level Clock Input Voltage	V _{CL}			0.6	V

V_{DD} and AV_{DD} = 5.0 V ±5%, T_{OP} = 0 to 70 °C, and the Power Supply Current is the sum of the average levels of terminals V_{DD} and AV_{DD}.

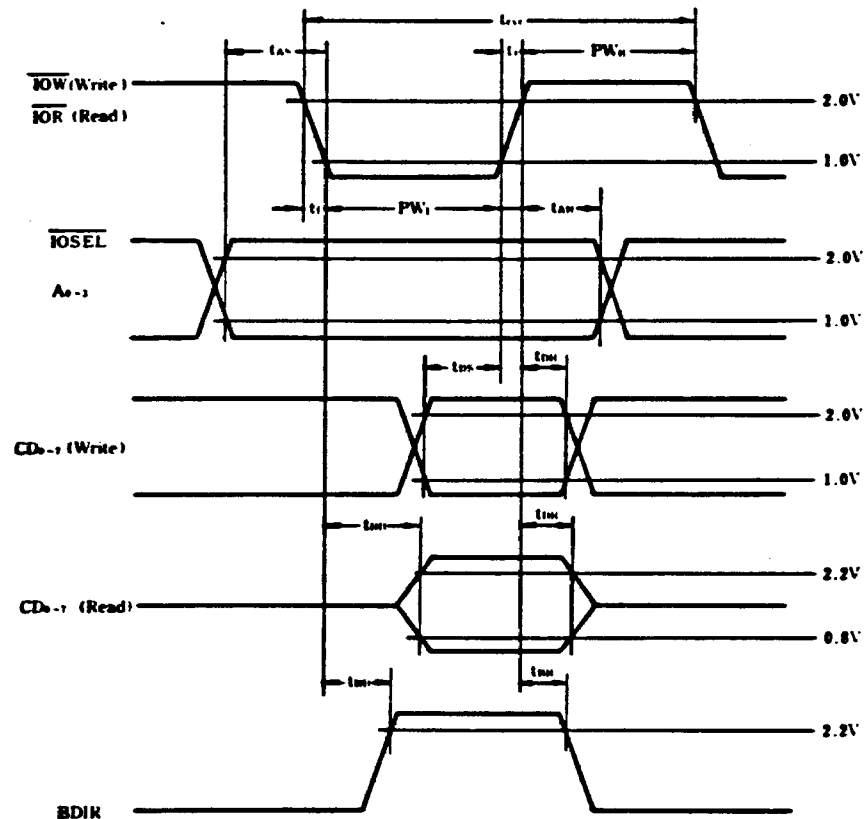
RL: A terminating resistor between Y, R, G, B, CH terminals and GND.

The Output Leakage Current (Tristate) applies to CD0 to CD7, RD0 to RD7, RA2/GD3 to RA0/GD1 at input condition; and to AD0 to AD12/*CAS, *LSEL, *HSEL, and MEMRDY when they are at high impedance.

(D) A.C. Characteristics (VDD=5.0V $\pm$ 5 %, TOP=0-70 °C)
D-1 I/O Register to CPU Interface

Parameter	Symbol	Minimum	Maximum	Unit
Cycle Time	tCYC	300		ns
Read/Write Low-Level Pulse Width	PWL	120		ns
Read/Write High-Level Pulse Width	PWH	120		ns
Read/Write Rise and Fall Time	tr, tf		25	ns
Address Setup Time	tAS	40		ns
Address Hold Time	tAH	10		ns
Data Setup Time	tDS	60		ns
Data Hold Time	tDH	10		ns
Data Delay Time	tDD		100	ns
Bus Direction Switching Hold Time	tBH	10		ns
Bus Direction Switching Delay Time	tBD		85	ns

$C_{CL} = 100 \text{ pF} + 2\text{LSTTL (CD0 to CD7, BDIR)}$



@CL = 100 pF + 2LSTTL (CD0 to CD7, BDIR, MEMRDY)

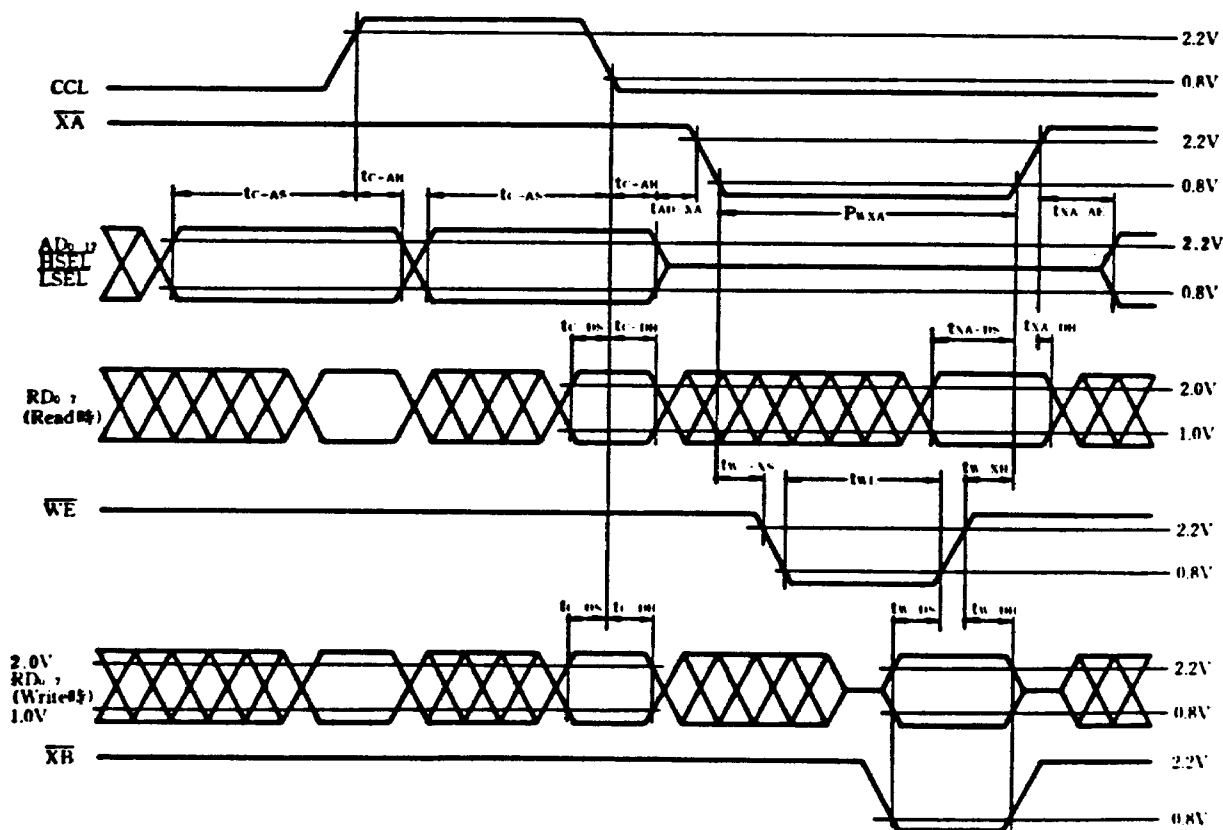


D-3 VRAM - LCDC Interface

(Master Clock = 21.47727 MHz, CL = 100 pF + 2LSTTL)

(1) SRAM Mode

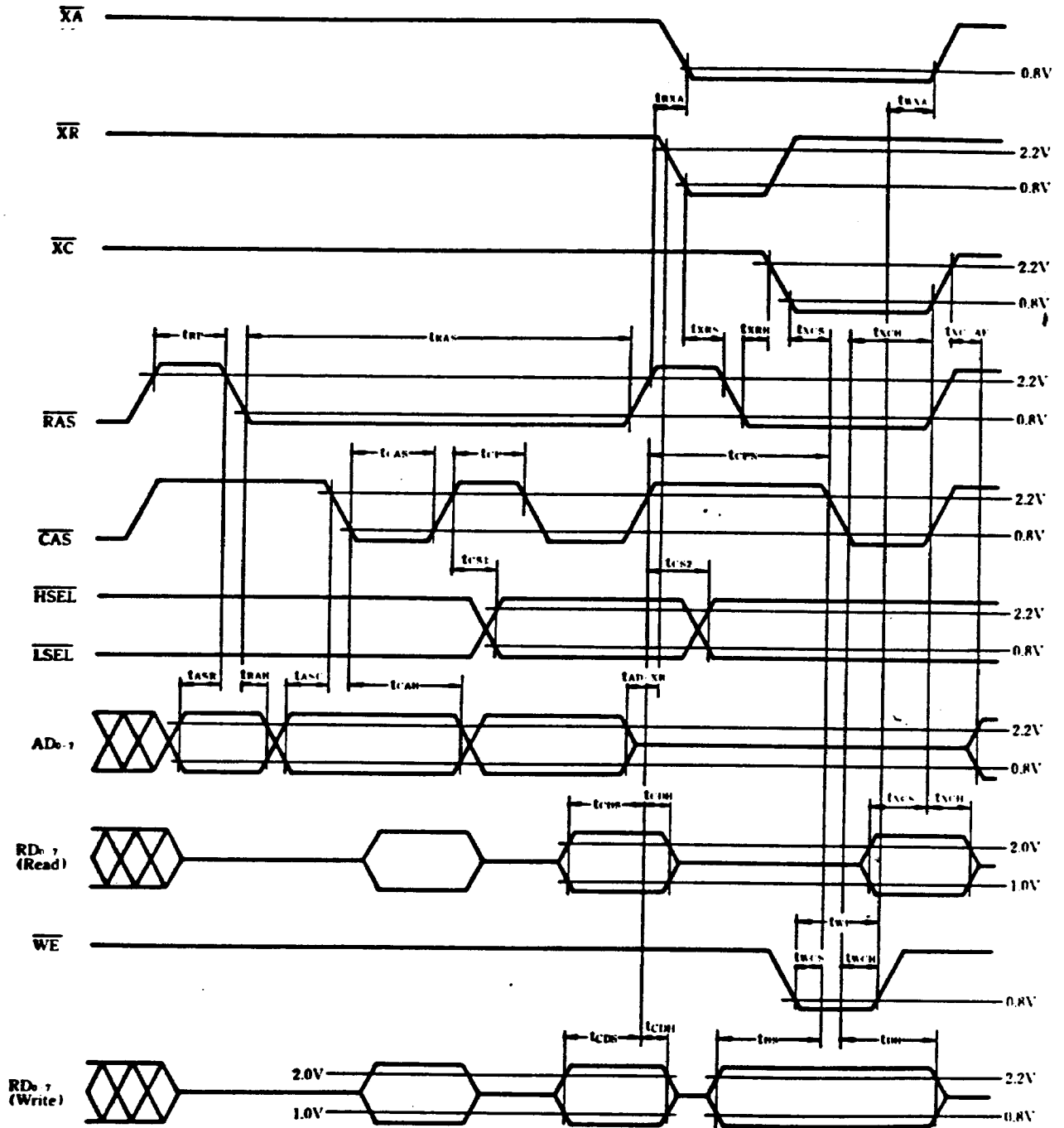
Parameter	Symbol	Minimum	Typical	Unit
CCL - Address Setup Time	tC-AS	150	163	ns
CCL - Address Hold Time	tC-AH	0	20	ns
CCL - Data Setup Time	tC-DS	0		ns
CCL - Data Hold Time	tC-DH	38		ns
Address Disable - *XA Delay Time	tAD-XA	0	20	ns
*XA - Address Enable Delay Time	tXA-AE	46	70	ns
*XA - Data Setup Time (During Read)	tXA-DS	50		ns
*XA - Data Hold Time (During Read)	tXA-DH	0		ns
*WE - *XA Setup Time (During Write)	tW-XS	23	35	ns
*WE - *XA Hold Time (During Write)	tW-XH	0	10	ns
*WE - Low-Level Pulse Width (During Write)	tWL	140	150	ns
*WE - Data Setup Time (During Write)	tW-DS	70	80	ns
*WE - Data Hold Time (During Write)	tW-DH	10		ns
*XA Low-Level Pulse Width	PWXA	210		ns



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(2) DRAM Mode

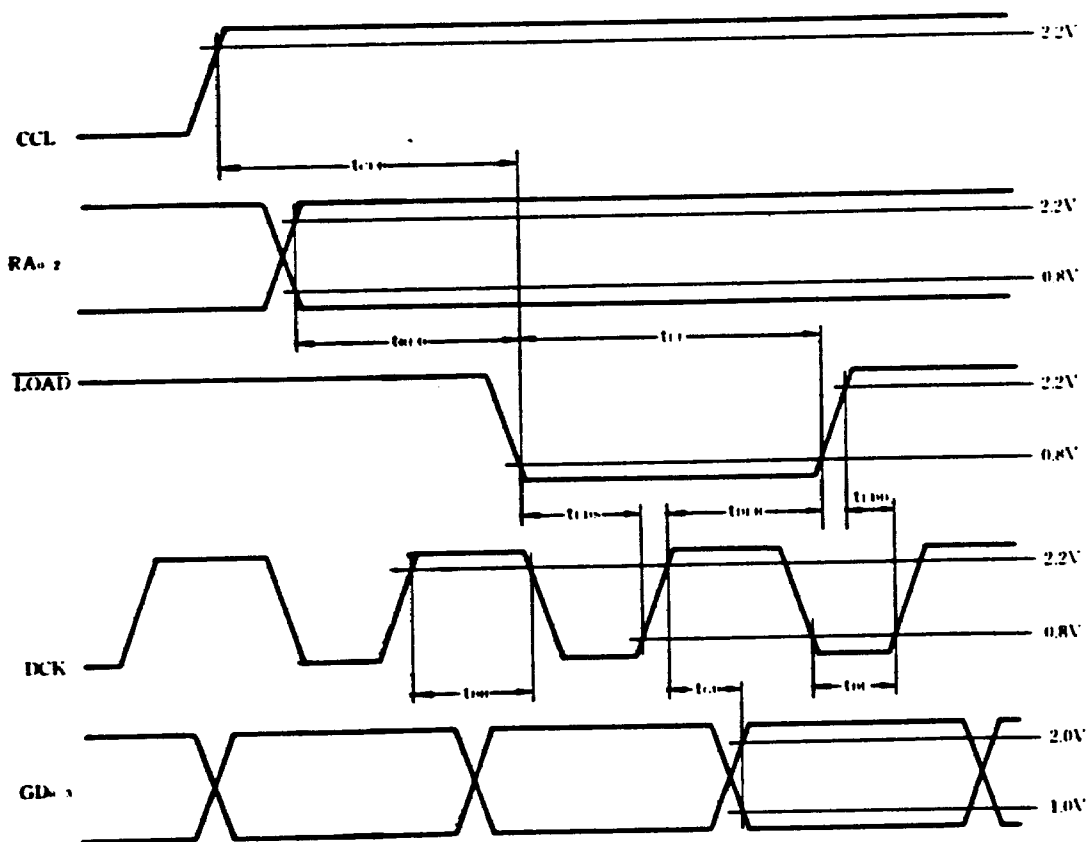
Parameter	Symbol	Minimum	Typical	Maximum	Unit
*RAS Precharge Time	tRP	98		108	ns
*RAS Pulse Width	TRAS	160		330	ns
*CAS Precharge Time (For other than Page Cycles)	tCPN	140	160		ns
*CAS Precharge Time (For Page Cycles Only)	tCP	60		75	ns
*CAS Pulse Width	tCAS	80		165	ns
Row Address Setup Time	tASR	23	40		ns
Row Address Hold Time	tRAH	20	30		ns
Column Address Setup Time	tASC	10		50	ns
Column Address Hold Time	tCAH	70		120	ns
*CAS - Data Setup Time	tCDS	0			ns
*CAS - Data Hold Time	tCDH	38			ns
*XC - Data Setup Time	tXCS	27			ns
*XC - Data Hold Time	tXCH	23			ns
Address Disable - *XR Delay Time	tAD-XR	23	45		ns
*XC - Address Enable Delay Time	tXC-AE	23	45		ns
*XR - *RAS Setup Time	tXRS	23	40		ns
*XC - *RAS Hold Time	tXRH	10	20		ns
*XC - *CAS Setup Time	tXCS	20	40		ns
*XC - *CAS Hold Time	tXCH	92	105		ns
*WE - *CAS Setup Time	tWCS	10	15		ns
*WE - *CAS Hold Time	tWCH	46	60		ns
*WE Pulse Width	tWP	70	80		ns
*CAS - Data Setup Time (During Write)	tDS	10			ns
*CAS - Data Hold Time (During Write)	tDH	92			ns
*RAS - *XA Delay Time	tRXA	23	45		ns
*WE - *XA Delay Time	tWXA	23	45		ns
*CAS - *SEL Delay Time - 1	tcS1	0	20		ns
*CAS - *SEL Delay Time - 2	tcS2	23	45		ns



D-4 GD - LCDC Interface

(Master Clock = 21.47727 MHz, CL = 100 pF + 2LSTTL)

Parameter	Symbol	Minimum	Typical	Maximum	Unit
CCL - LOAD Delay Time	tCLD	180			ns
RA - LOAD Delay Time	tRLD	300			ns
LOAD - DCK Delay Time	tLDD	10			ns
DCK - LOAD Hold Time	tDLH	30			ns
LOAD - DCK Setup Time	tLDS	25			ns
GD Delay Time	tGD			25	ns
DCK High-Level Pulse Width	tDH	30			ns
DCK Low-Level Pulse Width	tDL	10			ns
LOAD Low-Level Pulse Width	tLL	70			ns

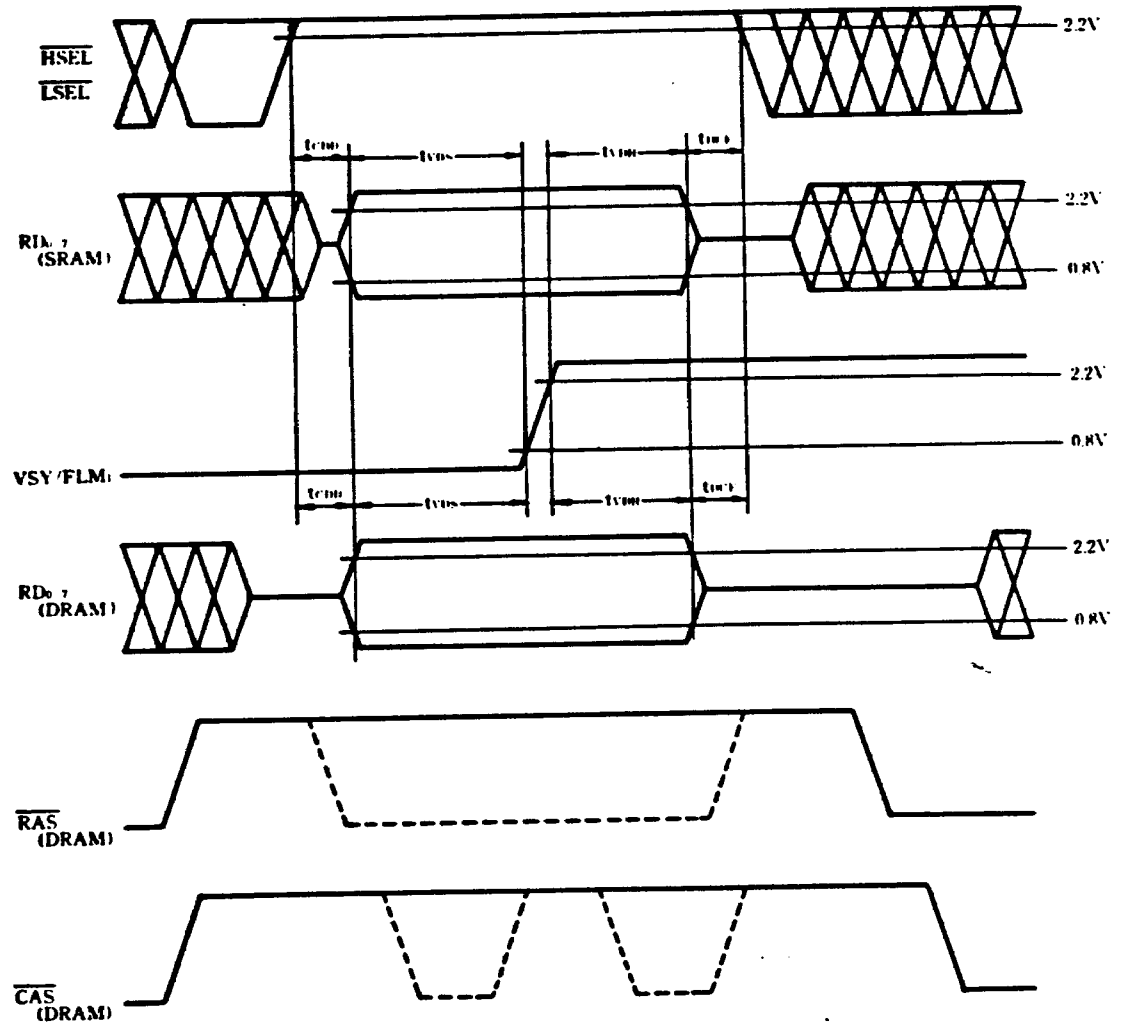


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D-5 Control Data Interface

(Master Clock = 21.47727 MHz, CL = 100 PF + 2LSTTL)

Parameter	Symbol	Minimum	Typical	Maximum	Unit
Chip Disable - Data Delay Time	tCDD	20	45		ns
Data - Chip Enable Delay Time	tDCE	20	45		ns
VS _Y - Data Setup Time	tVDS	120	160		ns
VS _Y - Data Hold Time	tVDH	120	160		ns

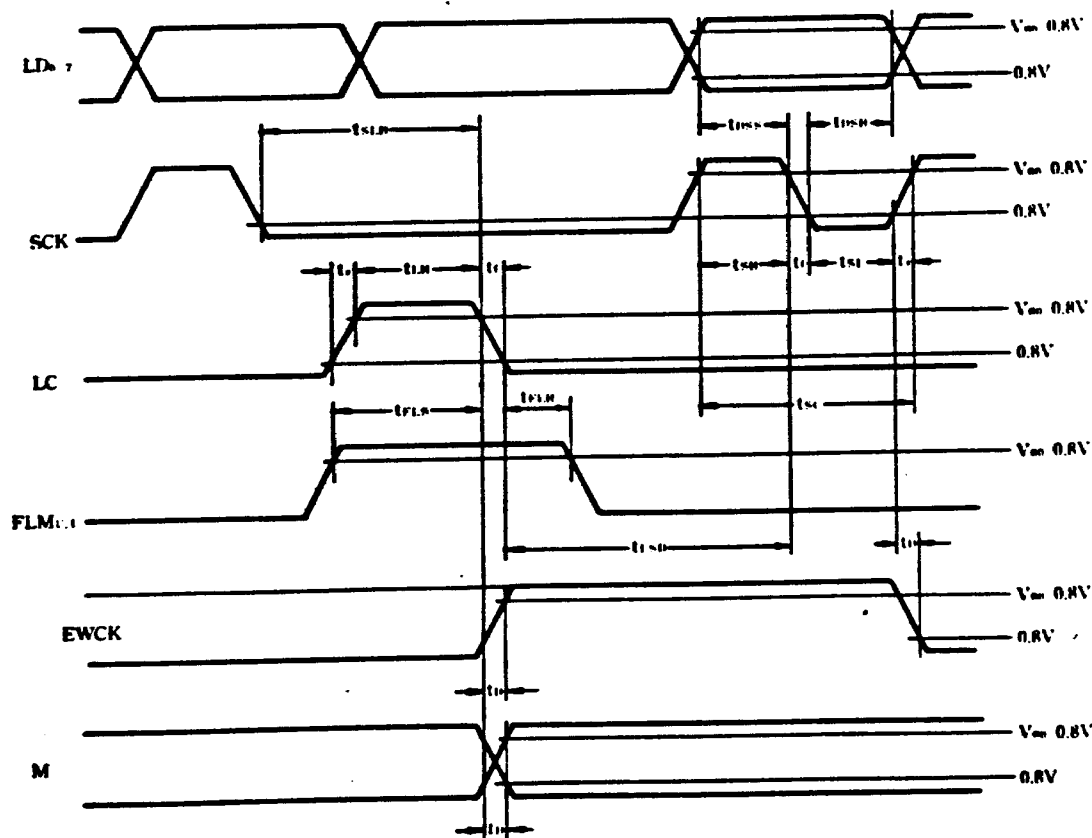


Note: The dotted lines for *RAS and *CAS are the normal timings. When using the Control Data Interface, *RAS and *CAS remain high.

D-6 LCD Interface (Master Clock = 21.47727 MHz, CL = 100 pF)
(Capacitive Load Only)

() Typ1, Typ3 Modes

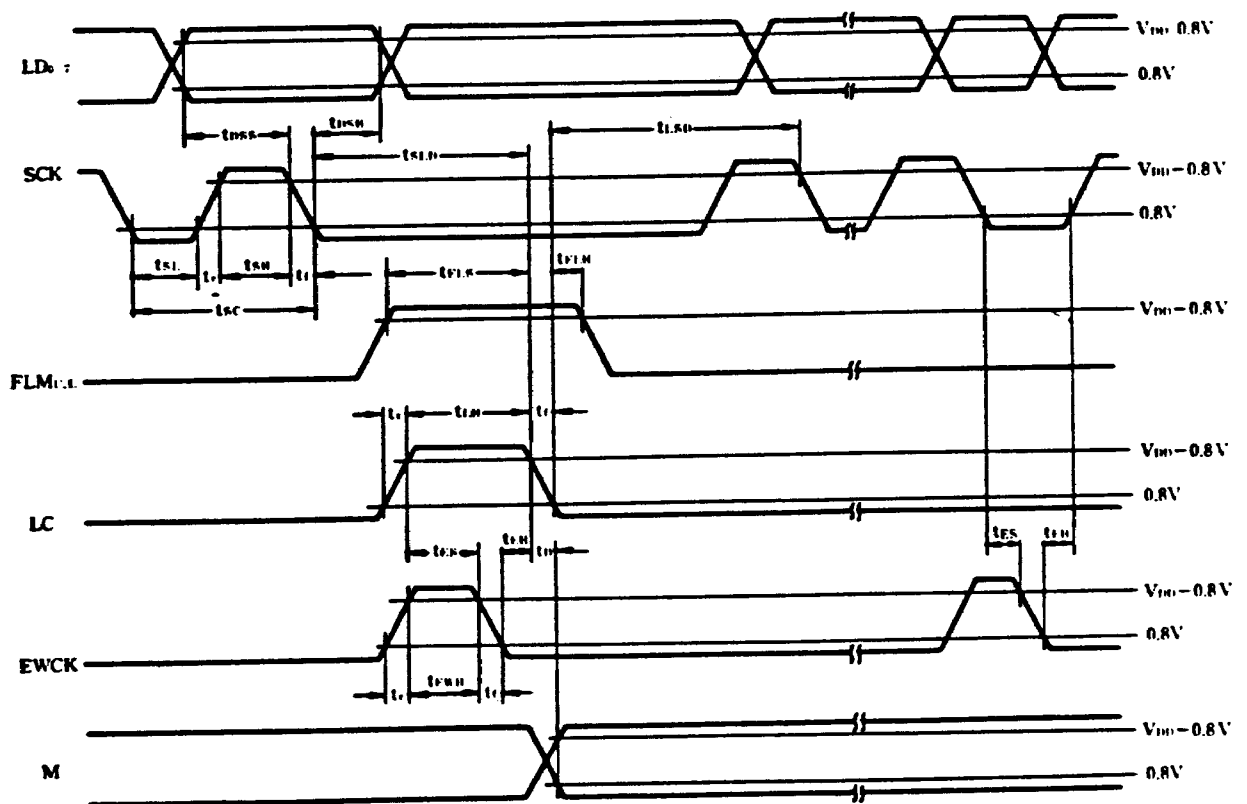
Parameter	Symbol	Minimum	Typical	Maximum	Unit
FLM - LC Setup Time	tFLS	320			ns
FLM - LC Hold Time	tFLH	320			ns
LC High-Level Pulse Width	tLH	320			ns
LC - SCK Delay Time	tLSD	8			µs
SCK - LC Delay Time	tSLD	400			ns
Data - SCK Setup Time	tDSS	50			ns
Data - SCK Hold Time	tDSH	50			ns
LC and SCK Rise and Fall Times	tr, tf			50	ns
SCK High-Level Pulse Width	tSH	50			ns
SCK Low-Level Pulse Width	tSL	50			ns
SCK Cycle Time	tSC	180			ns
EWCK, M Delay Times	tD			50	ns



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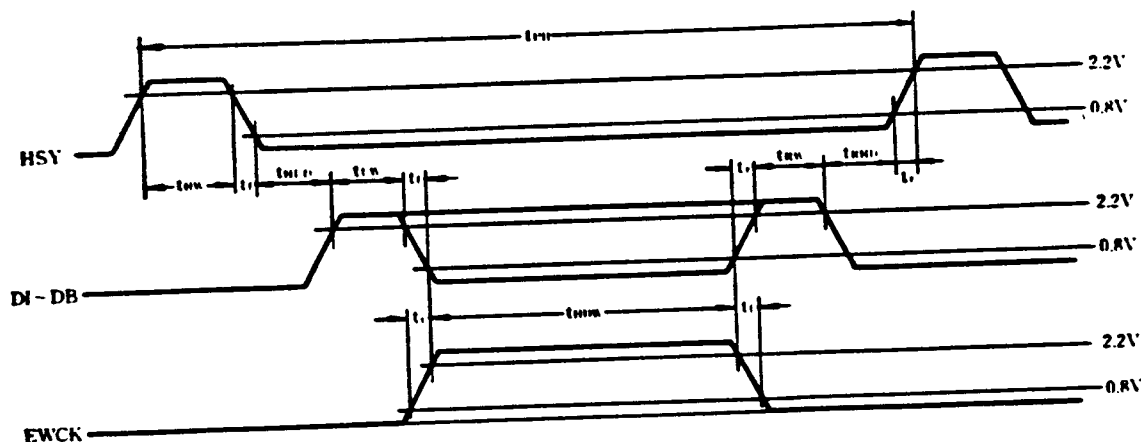
(2) Typ2 Mode

Parameter	Symbol	Min.	Typical	Max.	Unit
FLM - LC Setup Time	tFLS	320			ns
FLM - LC Hold Time	tFLH	320			ns
LC High-Level Pulse Width	tLH	320			ns
LC - SCK Delay Time	tLSD	8			μs
SCK - LC Delay Time	tSLD	500			ns
Data - SCK Setup Time	tDSS	140			ns
Data - SCK Hold Time	tDSH	140			ns
LC, SCK and EWCK Rise and Fall Times	tr, tf			50	ns
SCK High-Level Pulse Width	tSH	320			ns
SCK Low-Level Pulse Width	tSL	320			ns
SCK Cycle Time	tSC	740			ns
EWCK Setup Time	tES	140			ns
EWCK Hold Time	tEH	140			ns
EWCK High-Level Pulse Width	tEWH	140			ns
M Delay Time	tD			50	ns



D-7 CRT Interface (Master Clock = 21.47727 MHz, CL = 100 pF)
 (1) Color Monitor (IBM Color Monitor, Linear RGB Monitor)

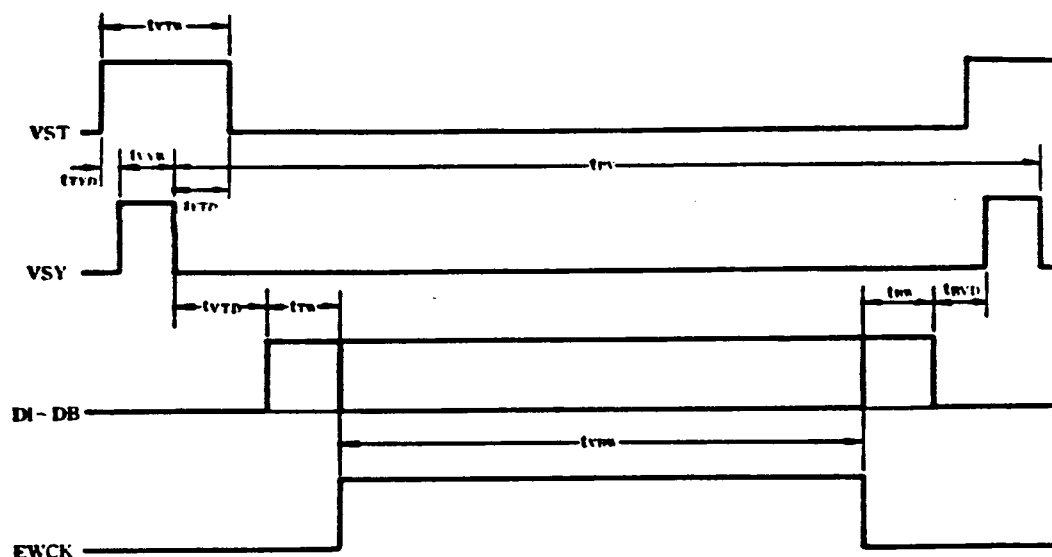
Parameter	Symbol	Minimum	Typical	Maximum	Unit
HSY High-Level Pulse Width	t _{HW}		4.5		μs
HSY - Left Border Delay Time	t _{HLD}	4.1		5.2	μs
Left Border width	t _{LW}	3.9		8.4	μs
Right Border width	t _{RW}	3.9		8.4	μs
Right Border - HSY Delay Time	t _{RHD}	1.5		2.5	μs
Horizontal Display Timing width	t _{HDW}	35.7		44.7	μs
HSY Interval	t _{PH}		63.695		μs
VSX Interval	t _{PV}		16.688		ms
VSX - Top Border Delay Time	t _{VTD}		764		μs
Top Border width	t _{TW}	1.4		1.7	ms
Bottom Border width	t _{BW}	1.3		1.8	ms
Bottom Border - VSX Delay Time	t _{BVD}		64		μs
Vertical Display Timing width	t _{VDW}	12.2		13.0	ms
VSX High-Level Pulse Width	t _{VYW}		196		μs
VST High-Level Pulse Width	t _{VTW}		1019		μs
VST - VSX Delay Time	t _{TYD}		56		μs
VSX - VST Delay Time	t _{YTD}		764		μs
Video Output Rise, Fall Times	t _r , t _f			50	ns



Notes: The timings for the R, G, and B outputs for a Linear RGB Monitor are the same as for DI to DB. The outputs, however, (offset voltage and amplitude) differ from DI and DB.

The Border width and Display Enable Timing width depend on the number of dots displayed.

For a PAL format, the VSP cycles are 20.0 ms, the Top Border time is increased by 1.7 ms, and the Bottom Border is increased by 1.5 ms.

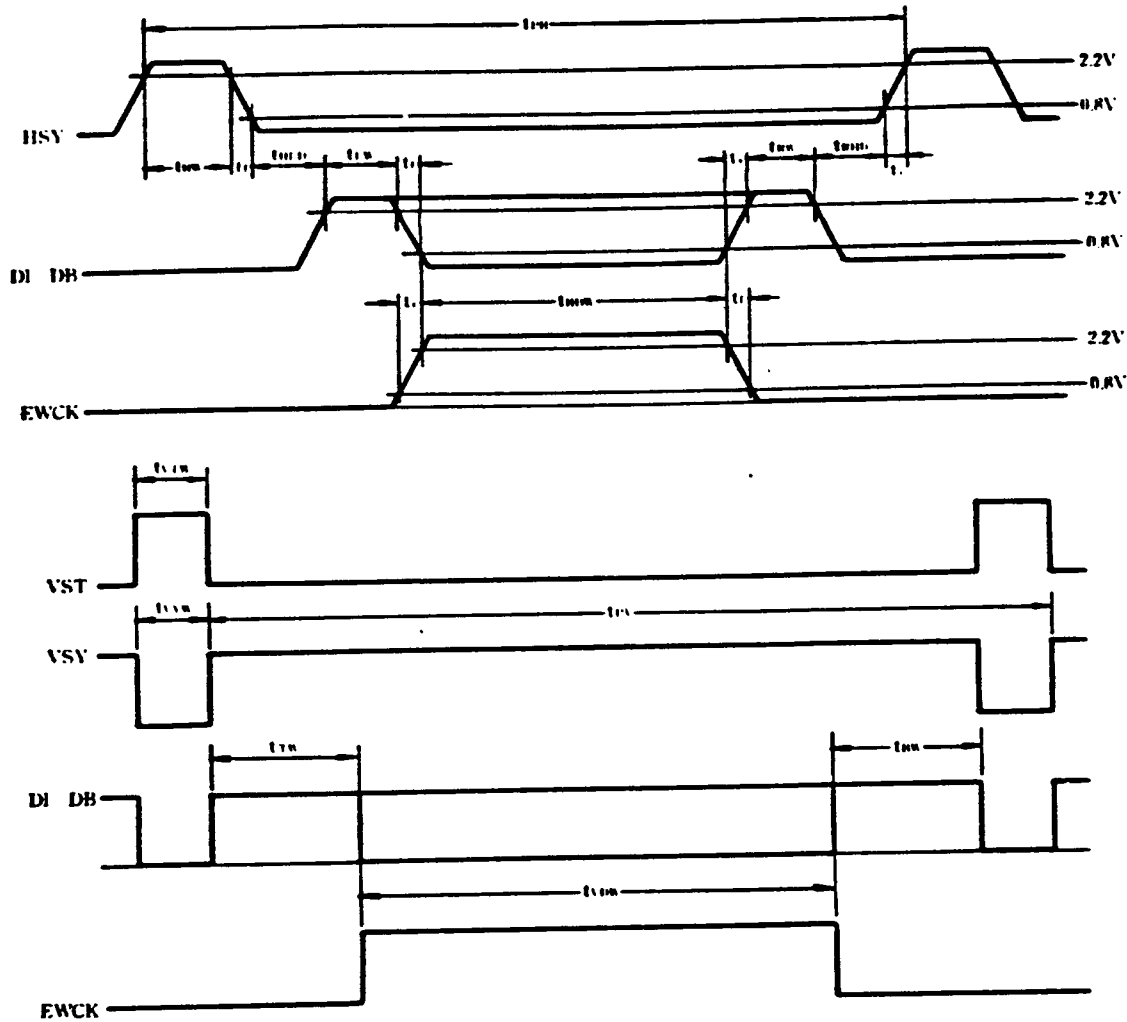


Note: DI to DB and EWCK are at the low level when near HSY.

(2) IBM Monochrome Monitor

Parameter	Symbol	Minimum	Typical	Maximum	Unit
HSY High-Level Pulse Width	tHW		8.9		μs
HSY - Left Border Delay Time	tHLD	-0.90		0.12	μs
Left Border width	tLW	0.55		5.03	μs
Right Border width	tRW	0.55		5.03	μs
Right Border - HSY Delay Time	tRHD	-0.12		0.90	μs
Horizontal Display Timing width	tHDW	35.8		44.7	μs
HSY Interval	tPH		54.76		μs
VSX Interval	tPV		20.04		ms
Top Border width	tTW	4.3		4.5	ms
Bottom Border width	tBW	3.7		4.2	ms
Vertical Display Timing width	tVDW	10.5		11.2	ms
VSX High-Level Pulse Width	tVYW		876		μs
VST High-Level Pulse Width	tVTW		876		μs
Video Output Rise, Fall Times	tr, tf			50	ns

Notes: The Border width and Display Enable Timing width depend on the number of dots displayed.



Note: DI to DB and EWCK are at the low Level when near HSY.

(3) Linear RGB Output Level (RL = 5.6 kOhms)

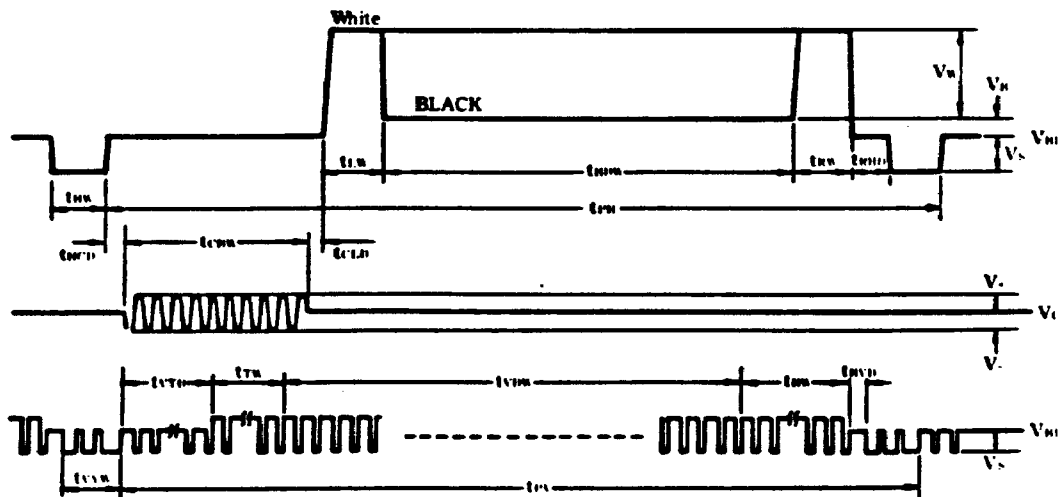
Parameter	Symbol	Minimum	Typical	Maximum	Unit
Offset Voltage	VO	1.1	1.3	1.7	V
Maximum Amplitude	VA	1.3	1.45	1.6	VP-P
Step Voltage	VS	0.14	0.21	0.27	V

Notes: RL is a terminating resistor between the R, G, and B terminals and GND.

(4) Composite Monitor (Terminating resistor for Y and CH = 5.6 kOhms)

Parameter	Symbol	Minimum	Typical	Maximum	Unit
HSY Pulse Width	t _{HW}		4.5		μs
HSY - Color Burst Delay Time	t _{HCD}	0.8		1.3	μs
Color Burst width	t _{CBW}		2.7		μs
Color Burst - Left Border Delay Time	t _{CLD}	0.6		1.2	μs
Left Border width	t _{LW}	3.9		8.4	μs
Right Border width	t _{RW}	3.9		8.4	μs
Right Border - HSY Delay Time	t _{RHD}	1.7		2.8	μs
Horizontal Display Timing width	t _{HDW}	35.7		44.7	μs
VSY - Top Border Delay Time	t _{VTP}		764		μs
Top Border width	t _{TW}	1.4		1.7	ms
Bottom Border width	t _{BW}	1.3		1.8	ms
Bottom Border - VSY Delay Time	t _{BVD}		64		μs
Vertical Display Timing width	t _{VDW}	12.2		13.0	ms
HSY Interval	t _{PH}		63.695		μs
VSY Interval	t _{PV}		16.688		ms
VSY Pulse Width	t _{VYW}		196		μs
Blanking Level	V _{BL}	1.2	1.5	1.9	V
(Black Level) - (Blanking Level)	V _B	70	85	100	mV
(SYNC Level) - (Blanking Level)	V _S	-0.35	-0.38	-0.42	V
(White Level) - (Black Level)	V _W	0.99	1.05	1.12	V
CH Offset Level	V _C	1.65	1.95	2.35	V
Burst + Side Level	V ₊	0.26	0.29	0.33	V
Burst - Side Level	V ₋	-0.26	-0.29	-0.33	V
CH + Side Maximum	C ₊	0.65	0.7	0.76	V
CH - Side Maximum	C ₋	-0.67	-0.72	-0.78	V

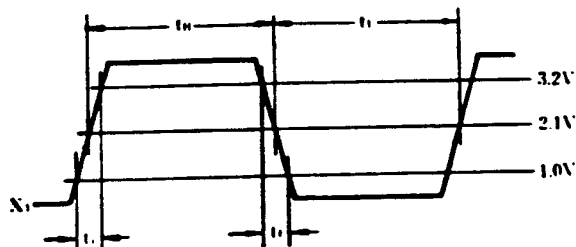
Notes: The White level is when the RGB = 777, and the maximum amplitude of CH is when RGB = 770.



D-8 Clock

(1) External Clock

Parameter	Symbol	Minimum	Typical	Maximum	Unit
Clock Frequency	f_c	18.6	21.47727	21.7	MHz
Clock Rise and Fall Time	t_r, t_f	47.5	50	6	ns
Clock Duty	Duty			52.5	%

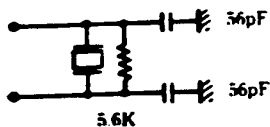


$$\text{Duty} = \frac{t_H}{t_H + t_L} \times 100\%$$

Notes: An external clock is input through the X1 terminal. The X2 terminal will not be connected (N.C.).

(2) Internal Clock

Parameter	Symbol	Minimum	Typical	Maximum	Unit
Clock Oscillation Frequency	fOSC		21.477		MHz



Using a 21.47727 MHz crystal

(3) Terminal Capacities

Parameter	Symbol	Condition	Minimum	Maximum	Unit
Input Terminal	CI			8	pF
Output Terminal	CO	Non-loaded		10	pF
Input/Output Terminal	CI/O	Non-loaded		12	pF
Output Load Capacity	CL			100	pF

Technical drawing of a 100-pin Quad Flat Pack (P-0.65TYR) showing top, side, and bottom views with dimensions.

Top View Dimensions:

- Pin pitch: $P=0.65TYR$
- Pin count: $100-0.3 \pm 0.1$
- Pin width: 0.15 ± 0.05
- Pin height: 2.15
- Pin spacing: 20.0 ± 0.2
- Pin array width: 24.3 ± 0.4
- Pin array height: 2.15
- Pin array length: 14.0 ± 0.2
- Pin array width: 18.3 ± 0.4
- Pin array length: 0.9 ± 0.2
- Pin array width: 1.95 ± 0.2
- Pin array length: 0.1 ± 0.05

Side View Dimensions:

- Pin height: 2.15
- Pin array width: 24.3 ± 0.4
- Pin array length: 14.0 ± 0.2
- Pin array width: 18.3 ± 0.4
- Pin array length: 0.9 ± 0.2
- Pin array width: 1.95 ± 0.2
- Pin array length: 0.1 ± 0.05

Bottom View Dimensions:

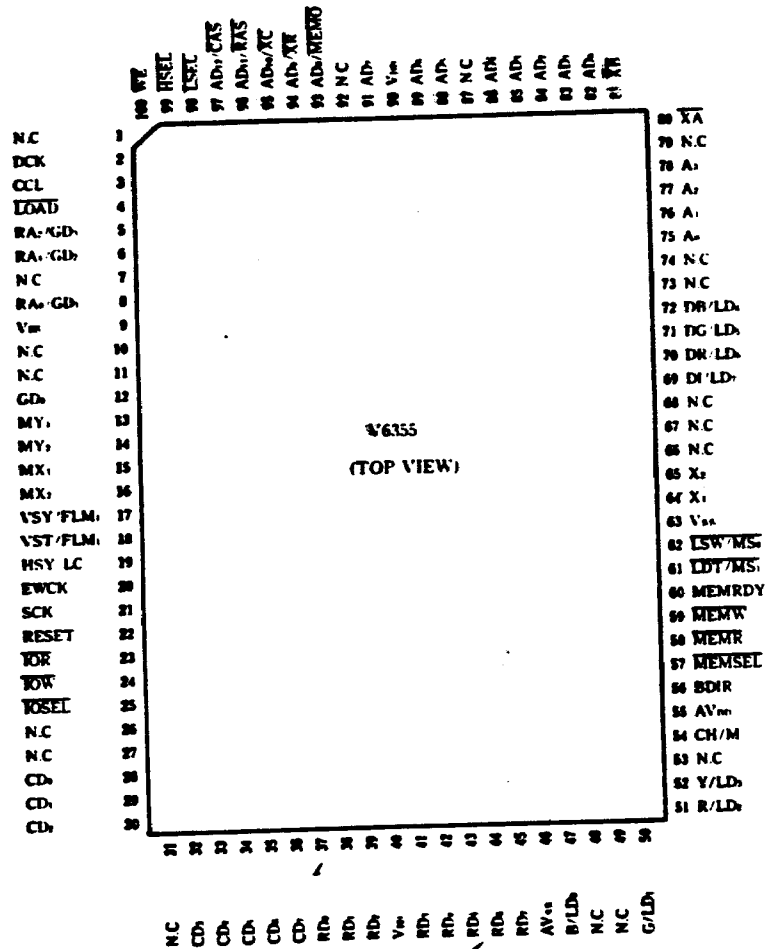
- Pin pitch: $P=0.65TYR$
- Pin count: $100-0.3 \pm 0.1$
- Pin width: 0.15 ± 0.05
- Pin height: 2.15
- Pin spacing: 20.0 ± 0.2
- Pin array width: 24.3 ± 0.4
- Pin array height: 2.15
- Pin array length: 14.0 ± 0.2
- Pin array width: 18.3 ± 0.4
- Pin array length: 0.9 ± 0.2
- Pin array width: 1.95 ± 0.2
- Pin array length: 0.1 ± 0.05

Scale: 1:1

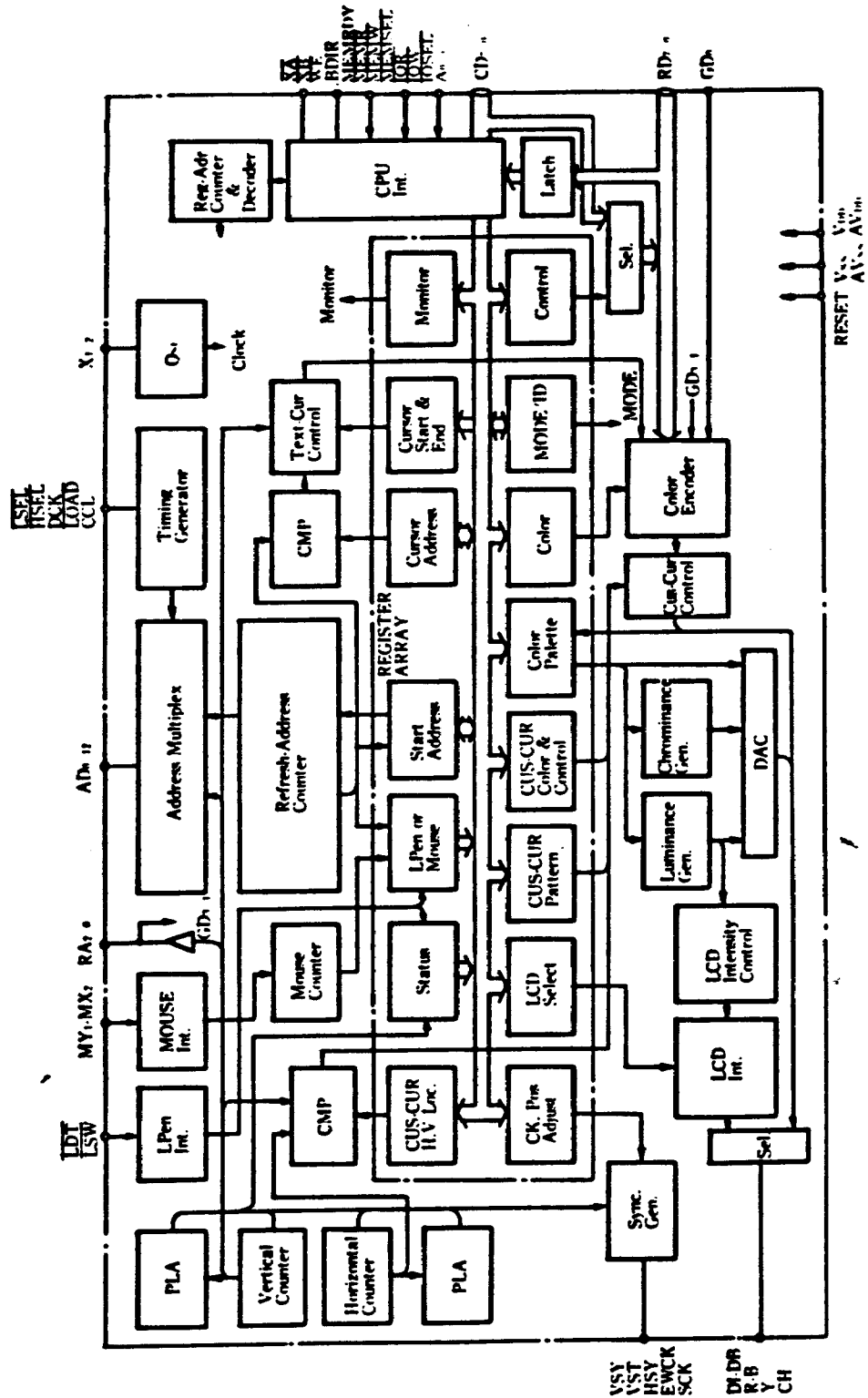
Scale : mm

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8. Pin Diagram



9. Block Diagram



10. Pins and their functions

Signal Name	Pin No.	I/O	Function
*IOR	23	I	Controls Read from the I/O Registers, Active low
*IOW	24	I	Controls Write to the I/O Registers, Active low
*IOSEL	25	I	Decode signal for upper bit of address signal of the I/O Registers, Selected when low
A0	75	I	I/O Register Address Signal
A1	76	I	
A2	77	I	
A4	78	I	
*MEMSEL	57	I	Decode signal for upper bits of address signal of memory addresses, Selected when low
*MEMR	58	I	Controls Read from memory, Active low
*MEMW	59	I	Controls Write to memory, Active low
MEMRDY	60	O	Memory (VRAM) address ready signal, Busy when low
*WE	100	O	Memory Write Enable
BDIR	56	O	Controls the (CPU Data Bus) Bidirectional Buffer
RESET	22	I	Reset Signal
CD0	28	I/O	Data bus for CPU
CD1	29	I/O	
CD2	30	I/O	
CD3	32	I/O	
CD4	33	I/O	
CD5	34	I/O	
CD6	35	I/O	
CD7	36	I/O	
RD0	37	I/O	VRAM Data bus
RD1	38	I/O	
RD2	39	I/O	
RD3	41	I/O	
RD4	42	I/O	
RD5	43	I/O	
RD6	44	I/O	
RD7	45	I/O	
AD0	82	O	VRAM Address
AD1	83	O	
AD2	84	O	
AD3	85	O	
AD4	86	O	
AD5	88	O	
AD6	89	O	
AD7	91	O	

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Signal Name	Pin No.	I/O	Function	
			SRAM	DRAM
AD8/*MEMO	93	O	VRAM Address	Memory Read Enable (*MEMR + *MEMSEL)
AD9/*XR	94	O	"	CPU Row Address Enable
AD10/*XC	95	O	"	CPU Column Address Enable
AD11/*RAS	96	O	"	*RAS
AD12/*CAS	97	O	"	*CAS
*XB	81	O	Memory Read/Write Data Enable	- - - - -
*XA	82	O	Memory Read/Write Timing (CPU Address Enable for SRAM)	
			SRAM (8-bit CPU)	16-bit CPU
*LSEL	98	O	Selects 1st 8K bytes	Selects Low byte
*HSEL	99	O	Selects 2nd 8K bytes	Selects High byte
			A/N Mode	640 x 200 Color Mode
RA2/GD3	5	I/O	Raster Address(Output)	Graphic Data (Input)
RA1/GD2	6	I/O	"	"
RA0/GD1	8	I/O	"	"
GD0	12	I	Dot data for Character Display	- - - - -
			CRT	LCD
VSF/FLML	17	O	Vertical Synchronize Signal	FLM of lower block
VST/FLMU	18	O	Vert. Retrace Timing	FLM of upper block
HSY/LC	19	O	Horizont. Synchronize Signal	Latch Clock
EWCK	20	O	Display Active Timing	Driver Enable Signal or Multiple-Weight Signal
SCK	21	O	- - - - -	Shift Clock
CH/M	54	O	Chrominance Signal	Signal for A.C. Driving
Y/LD3	52	O	Intensity (Black/White Composite) Signal	Upper Block Output Data
R/LD2	51	O	Linear RGB Output	"
G/LD1	50	O	"	"
B/LD0	47	O	"	"
DI/LD7	69	O	IBM Monitor Output	Lower Block Output Data
DR/LD6	70	O	"	"
DG/LD5	71	O	"	"
DB/LD4	72	O	"	"

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Signal Name	Pin No.	I/O	Function
DCK	2	O	Dot Shift Clock (For PS)
CCL	3	O	VRAM Data Latch Clock
*LOAD	4	O	P/S Load Signal
MY1	13	I	Mouse Y-direction Rotational Pulse
MY2	14	I	"
MX1	15	I	Mouse X-direction Rotational Pulse
MX2	16	I	"
*LDT/*MS1	61	I	Lightpen Detect Signal or Mouse Switch Signal
*LSW/*MS0	62	I	Lightpen or Mouse Switch Signal
X1	64	I	Crystal Oscillator, External Clock Input to X1
X2	65	O	"
AVSS	46	I	0 V Power for Analog circuits (DAC)
AVDD	55	I	+5 V "
VSS	9	I	0 V Power for Digital circuits
VSS	63	I	0 V "
VDD	40	I	+5 V "
VDD	90	I	+5 V "

The specifications of this product are subject to improvement changes without prior notice.

AGENCY

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